

PROJECT TRICORDER Mk I

Field operations manual · Revision 0.6.47

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RENT-A-HAL Android Edition, implemented as Project Tricorder Mk I.

1. Warnings

NAVIGATION. Not certified by FAA, NASA, EASA, or ICAO. Horizon, G-meter, GPS speed, baro altitude proxy, satellites, stars, SOL, moon, GNSS clock, HF forecasts, and walking NAV **MUST NEVER** be used to fly, land, or navigate any craft or vehicle. Eyeballs own fences and traffic. This load is not DO-178C airborne software.

WATCH. Weapon and gunfire phrases are caption keywords and YAMNet labels. Not a weapons detector. Not a 911 substitute.

COMLINK. Nearby mesh is a pocket-range landing-party radio. It is not a secure tactical network, not a 911 path, and not encrypted end-to-end in this revision. Treat PTT as a shouted note across the camp.

MEDICAL / MEMORY ALPHA. Encyclopedia briefings, not a clinician. MEMORY ALPHA is Simple English Wikipedia, not the Star Trek wiki.

MARKS. Not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang.

2. What this unit is

Mk I is an integration. The APK is the face. Four Termux servers on loopback (chat :8080, vision :8081, image :8188, tts :5000) plus on-device TFLite packs are the instrument. Comlink rides Google Nearby Connections inside the APK — not a Termux Matrix/SIP switchboard. Reference computer: Galaxy Z Fold 3, Snapdragon 888, 12 GB.

3. Comlink (0.6.46–0.6.47)

3.1 Why not Dendrite or Asterisk

A homeserver on every Fold 3 needs names, TLS, and a network that survives a valley. SIP is for live voice streams. Landing-party traffic is JSON events and a one-shot PTT note. Nearby P2P_CLUSTER already in the APK is the right radio. Matrix can wait for an optional Earth-link store-and-forward later.

3.2 Identity

GUID is TC- plus ten hex characters from a SHA-256 of ANDROID_ID. Not IMEI (restricted, often empty). Not a Wi-Fi MAC (randomized). Callsign is a sticker you assign: Spock, Kirk, UNIT by default. Advertise name is CALLSIGN|GUID.

3.3 Automatic link

Comlink starts when the face finishes loading. Units advertise and discover on service com.projecttricorder.comlink. The lexicographically smaller GUID requests the handshake so both sides do not collide. Accept is automatic. Yellow lamp = heard (reachable). Green = linked. Grey = seen earlier, now quiet.

3.4 Operator pane

Say TRICORDER COMLINK, TRICORDER COM LINK, TRICORDER COMMLINK, or bare COMLINK / COMMLINK. Or tap Comlink on the nameplate. The pane shows assigned-to name, GUID, and the roster. SAVE NAME writes the callsign (18 characters) and re-advertises. Touch a roster name to mute or unmute that unit's alerts; the radio stay stays up. PTT on a row: speak; the far tube plays robot voice "Beep beep beep. From Joe. ..." Always robot voice for device-to-device. Local watches may still use Kokoro.

3.5 Shared alerts

A local context/watch hit is broadcast as JSON {kind:context, from, guid, source, text, hhmm}. The far unit speaks "Remote tricorder Spock detected gunfire ..." and logs it. Muted GUIDs are dropped on receive. Both units must run 0.6.46 or newer. Bluetooth and Nearby Wi-Fi permission on. Range is pocket radio, not a ridgeline.

3.6 Speech aliases (0.6.47)

Android STT often emits tricorder and commlink as single tokens. Unknown verbs recited the H/F menu. 0.6.47 maps tricorder / trick order / commlink / com link / comms onto the same doors.

4. Walking NAV (0.6.38–0.6.45)

H cycles off → horizon → NAV → off. Line-of-sight walking only. First NAV open with no stored camp marks CAMP at the current fix. CAMP key or TRICORDER CAMP marks home here. BACKTRACK points DEST at CAMP. Dest forms: 41.50N 74.01W ; N 41.50 W 74.01 ; 41.50, -74.01 ; plus code ; ///w3w (key required). Bare 41.50, 74.01 is east longitude. Spoken decimals rebuilt from 41 50 N 74 01 W.

Coordinate matrix sits under STANDBY (CAMP / YOU / DEST / SAID). The CRT pane scrolls: matrix, horizon, map, GO/MARK/CAMP/BACKTRACK. Swipe the tube. Off-track chime beyond 30 m. GPS stale > 8 s: dead-reckon along heading, labeled DR. 0.6.42 declared const strip twice and blanked the face; do not sideload it.

5. Face map

I Inquiry · II Sensor · III Sentry (continuous) · IV Audio · T/XLATE · H Horizon/NAV · S Sat/Stars · O Sol (space, earth weather, moon) · L E V P logs · IPL/ALL abort · STT kill speech. ROBO: dark Kokoro · green robot · yellow H/F · red MUTE.

6. Hands-free menu

Yellow [H/F R]: TRICORDER then SENSOR, AUDIO, SENTRY, TRANSLATOR, HORIZON, NAV, SATELLITES, SOL, CLOCK, DICTATION, COMLINK, SETTINGS, BACK END, DIAGNOSTICS, SUMMARIZE READINGS, HELP, CAMP, BACKTRACK, NAVIGATE TO ..., MARK.

7. Detector, readings, flights

TOP 3 fused bars; HIGHLY LIKELY is the bottom bar. Cattle/moo is not cat. Readings drill by category; flights by tail and airframe, kind=flight only. Overhead card: OpenSky + photo + airframe facts + city dossier. DISMISS AIRFRAME / BACK TO AIRFRAME.

8. Clock, moon, places, dictation, watch

CLOCK: GNSS face, analog/digital, 12/24, labeled timers. Field novelty, not a time standard. Moon is Sol's third press. Places: city + plus code; optional what3words key.

Dictation: NEW / EDIT nnnn / LOG nnnn / SUMMARIZE / MINDMAP / SPEAK / KILL / ZULU
SAVE|ABORT / cursor words / PUNCTUATION PERIOD.

Sentry captions every dwell. Sure weapon: klaxon + WEAPON DETECTED ×3. Audio priority gunfire, scream, siren, traffic, animals, humans, entertainment. IPL aborts work in flight. Text over 80 characters is chunked for Kokoro.

9. Knowledge banks and backends

MEMORY ALPHA = Simple English Wikipedia. MEDICAL body from English Wikipedia. One-shot: memory alpha Airbus A320. Termux four-pack ~/bin/tricorder start on 8080/8081/8188/5000. YAMNet ships in the APK; BirdNET, flora/fauna, translator packs fetch on first Earth link.

10. Hardware

Reference: Galaxy Z Fold 3, Snapdragon 888, 12 GB. Onn-class tablets are not a host. Comlink needs Play services Nearby on both units.

11. Document control

Rev 0.6.47. On-device copy: assets/www/manual.html. This PDF is the field narrative through Comlink and walking NAV. Prior PDF: 0.6.45. Prior Word: 0.6.37.